Section 2: Selected exercise solutions

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Problem 1. *Curvature and torsion in an arbitrary parameter.*

Solution. Write $v = \|\dot{c}\|$. Since $\dot{c} = vT$ and $\frac{d}{dt} = v\frac{d}{ds}$, the Frenet equations give

$$\ddot{c} = \dot{v}T + v^2\kappa N.$$

Therefore

$$\dot{c} \times \ddot{c} = v^3\kappa B,$$

and hence

$$\boxed{\kappa = \frac{\|\dot{c} \times \ddot{c}\|}{\|\dot{c}\|^3}.}$$

The B-component of $\dddot{c}$ is $v^3\kappa\tau B$. Consequently,

$$\det(\dot{c}, \ddot{c}, \dddot{c}) = v^6\kappa^2\tau,$$

while

$$\|\dot{c} \times \ddot{c}\|^2 = v^6\kappa^2.$$

Thus

$$\boxed{\tau = \frac{\det(\dot{c}, \ddot{c}, \dddot{c})}{\|\dot{c} \times \ddot{c}\|^2}.}$$

For an oriented plane curve,

$$\boxed{k = \frac{\det(\dot{c}, \ddot{c})}{\|\dot{c}\|^3}.}$$

Problem 4. *A length-minimizing regular curve is a line segment.*

Solution. Assume $p \neq q$, and set

$$u = \frac{q - p}{\|q - p\|}.$$

For any regular curve $c : [a, b] \to \mathbb{R}^n$ from p to q, Cauchy–Schwarz gives

$$\begin{aligned} L(c) &= \int_a^b \|c'(t)\| \, dt \\ &\geq \int_a^b \langle c'(t), u \rangle \, dt \\ &= \langle q - p, u \rangle = \|q - p\|. \end{aligned}$$

The line segment attains this bound.

Equality in Cauchy–Schwarz requires

$$c'(t) = \lambda(t) u, \qquad \lambda(t) > 0.$$

Thus $c(t) = p + \mu(t) u$, where μ ranges from 0 to $\|q - p\|$. The image of c is therefore precisely the segment joining p to q.

Problem 5. *The tangent vectors of* $c(t) = (3t, 3t^2, 2t^3)$.

Solution. Draw the tangent vectors from the origin and denote their endpoints by

$$X(t) = c'(t) = (3, 6t, 6t^2).$$

Let

$$a = \frac{1}{\sqrt{2}}(1, 0, 1),$$

the unit vector along the line $x - z = y = 0$. Then

$$\langle X(t), a\rangle = \frac{3 + 6t^2}{\sqrt{2}},$$

and

$$\|X(t)\| = \sqrt{9 + 36t^2 + 36t^4} = 3(1 + 2t^2).$$

Hence

$$\frac{\langle X(t), a\rangle}{\|X(t)\|} = \frac{1}{\sqrt{2}}.$$

Every $X(t)$ makes the constant angle $\pi/4$ with the axis spanned by a. Thus the endpoints lie on the circular cone with that axis and half-angle $\pi/4$.

Problem 7. *A plane curve with $k(s) = s^{-1/2}$.*

Solution. For $s > 0$, the tangent angle satisfies

$$\varphi'(s) = k(s) = s^{-1/2}.$$

Up to a rotation,

$$\varphi(s) = 2\sqrt{s}, \qquad T(s) = \big(\cos(2\sqrt{s}), \sin(2\sqrt{s})\big).$$

Integrating $c' = T$ and setting $u = \sqrt{s}$ gives

$$\begin{aligned} x(s) &= \int 2u\cos(2u)\,du = u\sin(2u) + \frac{1}{2}\cos(2u), \\ y(s) &= \int 2u\sin(2u)\,du = -u\cos(2u) + \frac{1}{2}\sin(2u). \end{aligned}$$

Thus, up to a Euclidean transformation (rotation, reflection, translation):

$$\boxed{c(s) = \left(\sqrt{s}\sin(2\sqrt{s}) + \frac{1}{2}\cos(2\sqrt{s}), -\sqrt{s}\cos(2\sqrt{s}) + \frac{1}{2}\sin(2\sqrt{s})\right).}$$

Indeed, $c' = T$, so c has unit speed and

$$k = \varphi' = s^{-1/2}.$$

Problem 9. *Arclength and curvature in polar coordinates.*

Solution. Let
$$c(\varphi) = r(\varphi)e_r(\varphi),$$
where
$$e_r = (\cos\varphi, \sin\varphi), \qquad e_\varphi = (-\sin\varphi, \cos\varphi).$$
Since $e_r' = e_\varphi$ and $e_\varphi' = -e_r$,
$$c' = r'e_r + re_\varphi, \qquad c'' = (r'' - r)e_r + 2r'e_\varphi.$$
Therefore
$$\|c'\| = \sqrt{(r')^2 + r^2},$$
and
$$\boxed{L = \int_{\varphi_1}^{\varphi_2} \sqrt{(r')^2 + r^2}\, d\varphi.}$$

The frame (e_r, e_φ) is positively oriented, so
$$\det(c', c'') = 2(r')^2 - r(r'' - r) = 2(r')^2 - rr'' + r^2.$$
Using the plane-curvature formula from Problem 1,
$$\boxed{k(\varphi) = \frac{2(r')^2 - rr'' + r^2}{\left((r')^2 + r^2\right)^{3/2}}.}$$

Problem 10. *Curvature of the Archimedean spiral $r(\varphi) = a\varphi$.*

Solution. Here

$$r = a\varphi, \qquad r' = a, \qquad r'' = 0.$$

Substitution into the formula from Problem 9 gives

$$\begin{aligned} k(\varphi) &= \frac{2a^2 + a^2\varphi^2}{(a^2 + a^2\varphi^2)^{3/2}} \\ &= \boxed{\frac{\varphi^2 + 2}{|a|(1 + \varphi^2)^{3/2}}}. \end{aligned}$$

This assumes $a \neq 0$. For the usual convention $a > 0$, the absolute value may be omitted.

Problem 12. *The polar curve* $r = \cos(2\varphi)$.

Solution. Since

$$r' = -2\sin(2\varphi),$$

the speed satisfies

$$\|c'(\varphi)\|^2 = (r')^2 + r^2 = 4\sin^2(2\varphi) + \cos^2(2\varphi) > 0.$$

Thus the curve is regular, including when $r = 0$.

In complex notation,

$$c'(\varphi) = e^{i\varphi}\big(r'(\varphi) + ir(\varphi)\big).$$

The factor $e^{i\varphi}$ winds once. The ellipse

$$r' + ir = -2\sin(2\varphi) + i\cos(2\varphi)$$

winds twice counterclockwise about the origin. Hence

$$\boxed{U_c = 1 + 2 = 3.}$$

Moreover,

$$2(r')^2 - rr'' + r^2 = 8\sin^2(2\varphi) + 5\cos^2(2\varphi) > 0.$$

The signed curvature never changes sign, so

$$\boxed{\int_c k\,ds = 2\pi U_c = 6\pi.}$$

The total absolute curvature is also 6π.

Problem 16. *Spherical slope lines.*

Solution of the first part. For a slope line,

$$A = \frac{\tau}{\kappa}$$

is constant. Since $\tau \neq 0$, also $A \neq 0$. The spherical-curve criterion from Theorem 2.10(ii) is

$$\frac{\tau}{\kappa} = \left(\frac{\kappa'}{\tau\kappa^2}\right)'.$$

Set $u = \kappa^{-2}$. Since $\tau = A\kappa$,

$$\frac{\kappa'}{\tau\kappa^2} = \frac{\kappa'}{A\kappa^3} = -\frac{1}{2A}u'.$$

The spherical-curve criterion becomes

$$A = -\frac{1}{2A}u'',$$

or

$$u'' = -2A^2.$$

Therefore

$$u(s) = -A^2s^2 + Bs + C,$$

and hence

$$\boxed{\kappa^2(s) = \left(-A^2s^2 + Bs + C\right)^{-1}.}$$

Each step is reversible, proving the converse as well.

Problem 16, continued. *A spherical slope line cannot reach the north pole.*

After translating and rescaling, take the unit sphere and let the fixed slope direction v be its polar axis. Put

$$\alpha = \langle T, v \rangle.$$

This is constant. If the curve starts on the equator at $s = 0$, its height is

$$z(s) = \langle c(s), v \rangle = \alpha s.$$

If the curve reached the north pole, its tangent there would be orthogonal to v, forcing $\alpha = 0$. But then $z \equiv 0$, so the curve would remain on the equator. Thus it cannot reach the pole.

For $\alpha > 0$, write

$$c(s) = \left(\sqrt{1 - \alpha^2 s^2} \cos \theta(s), \sqrt{1 - \alpha^2 s^2} \sin \theta(s), \alpha s \right).$$

The unit-speed condition gives

$$\theta'(s)^2 = \frac{1 - \alpha^2 - \alpha^2 s^2}{(1 - \alpha^2 s^2)^2}.$$

Hence

$$\alpha^2 s^2 \leq 1 - \alpha^2.$$

At the endpoint equality holds and $\theta' = 0$. The tangent then has no component along the corresponding small circle, so the curve meets that circle orthogonally.

Problem 18. *The Darboux equations.*

Solution. Let
$$D = \tau e_1 + \kappa e_3.$$
Using the orientation relations
$$e_1 \times e_2 = e_3, \qquad e_2 \times e_3 = e_1, \qquad e_3 \times e_1 = e_2,$$
we obtain
$$\begin{aligned} D \times e_1 &= \kappa e_2, \\ D \times e_2 &= -\kappa e_1 + \tau e_3, \\ D \times e_3 &= -\tau e_2. \end{aligned}$$
These are exactly the Frenet equations
$$e_1' = \kappa e_2, \qquad e_2' = -\kappa e_1 + \tau e_3, \qquad e_3' = -\tau e_2.$$
Thus the Frenet equations are equivalent to
$$\boxed{e_i' = D \times e_i, \qquad i = 1, 2, 3.}$$

Problem 20. *Helices, slope lines, and the Darboux vector.*

Solution. Differentiating $D = \tau e_1 + \kappa e_3$ gives

$$\begin{aligned} D' &= \tau' e_1 + \tau\kappa e_2 + \kappa' e_3 - \kappa\tau e_2 \\ &= \tau' e_1 + \kappa' e_3. \end{aligned}$$

Therefore

$$D' = 0 \quad \Longleftrightarrow \quad \kappa' = \tau' = 0.$$

A Frenet curve has constant curvature and torsion precisely when it is a circular helix, including the planar circle as the case $\tau = 0$. Hence

$$\boxed{c \text{ is a helix} \iff D \text{ is constant.}}$$

Since

$$\frac{D}{\|D\|} = \frac{\tau e_1 + \kappa e_3}{\sqrt{\kappa^2 + \tau^2}},$$

a constant ratio τ/κ makes this a fixed vector. Conversely, if $D/\|D\|$ is fixed, it is everywhere orthogonal to e_2; Theorem 2.11 then implies that c is a slope line. Thus

$$\boxed{c \text{ is a slope line} \iff \frac{D}{\|D\|} \text{ is constant.}}$$

Problem 23. *The determinant of the derivatives of a Frenet curve in $\mathbb{R}^n$.*

Solution. With respect to the positively oriented Frenet frame,

$$c' = e_1.$$

Repeated differentiation of the Frenet equations shows inductively that

$$c^{(j)} = (\kappa_1\kappa_2 \cdots \kappa_{j-1})e_j \text{ a linear combination of } e_1, \ldots, e_{j-1}.$$

Thus the coordinate matrix of

$$c', c'', \ldots, c^{(n)}$$

in the Frenet frame is triangular. Its diagonal entries are

$$1, \quad \kappa_1, \quad \kappa_1\kappa_2, \quad \ldots, \quad \kappa_1\kappa_2 \cdots \kappa_{n-1}.$$

Since the Frenet frame has determinant 1,

$$\begin{aligned} \det(c', c'', \ldots, c^{(n)}) &= \prod_{j=2}^{n} \left(\kappa_1 \cdots \kappa_{j-1}\right) \\ &= \boxed{\prod_{i=1}^{n-1} \kappa_i^{n-i}}. \end{aligned}$$